LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Specification of hydraulic oil, need for clean pipelines and use of protective ointment while handling oils.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Specification of hydraulic oil, need for clean pipelines and use of protective ointment while handling oils.
5. Specific Objectives	17.1.2.1 Trainee will be able to describe typical hydraulic system and construction of its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of oil spill containment and application of protective ointment on skin; Safety: Wear gloves and eye protection; Supervision: Instructor monitors technique and safety compliance.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of hydraulic fluid in an aircraft system?
(b) Why must hydraulic pipelines be free from contaminants before installation?
(c) What type of injury can occur from direct contact with hydraulic oil?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Viscosity and temperature characteristics • Flash point and fire hazard limits • Anti-corrosion and anti-wear additives • Compatibility with seals and hoses • Standard specification grades for aircraft	L & Ds	PPT, IB	15 min
(b)	• Sources of contamination in pipelines • Impact of particulates on valve operation • Water contamination and freezing risks • Procedures for cleaning pipelines during maintenance • Inspection methods for pipeline cleanliness	L & Ds	PPT, PDF	20 min
(c)	• Purpose of protective ointment on skin • Correct application technique for handlers • Removal procedures after handling • Storage and disposal of used ointment • Safety precautions during ointment application	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain why hydraulic oil specifications vary between different aircraft systems.
(b) Describe the consequences of using oil with incorrect viscosity in a hydraulic system.
(c) List three reasons why clean pipelines are critical for hydraulic system reliability.

12. Home Assignment and Follow Up	Prepare a brief report comparing the properties of hydraulic oil used in LP 02 versus current system requirements, highlighting differences in specification.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Brief description, purpose of a typical reservoir, EDP, hand pump, NRV pressure relief valve, automatic cut out valve, thermal relief valve, sequence valve, throttling valve, accumulator jack selector valve and filters	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min